

Mathematics and Computers in Simulation.
Report on the manuscript MATCOM-D-26-01427.

Title: *High-Accuracy Numerical Solutions of Particle Motion in Static Magnetic Fields*

In this work, the authors investigate the Parker-Sochacki (PS) method for the numerical integration of charged-particle trajectories in static magnetic fields. The method is compared with RK4, RK45, and, in the dipole-field test case, with a Gauss-Legendre Runge-Kutta (RKG) method. The numerical results suggest that the PS method can provide very accurate long-time simulations, especially in terms of kinetic-energy conservation.

In my opinion, the topic is interesting and in line with the scope of the journal. The numerical experiments are promising. However, I believe that the manuscript requires major revision before it can be considered for publication. My main comments are listed below.

1. The PS method for ODEs is already known. The original contribution of the paper seems to be the application of PS to long-time charged-particle motion in selected static magnetic fields, together with numerical comparisons. This should be stated in a clear way.
2. The literature review should mention some standard methods specifically designed for charged-particle integration. In particular, the Boris algorithm [C. K. Birdsall and A. B. Langdon, *Plasma Physics via Computer Simulation*] is a classical reference method in plasma physics and would provide a useful point of comparison.
3. More generally, the comparison should include or at least discuss specialized geometric methods for charged-particle dynamics. Comparing only with RK4, RK45, and RKG gives a partial view of the available numerical methods.
4. The theoretical part mainly consists of deriving the PS recurrence relations for the magnetic-field configurations considered. This is useful and necessary for the implementation. However, the manuscript does not provide much theoretical analysis of the resulting numerical method, for example regarding the global error, or the effective order of accuracy after truncation and tethering. Since several conclusions concern long-time stability and efficiency, I think that these points should be supported by some additional analysis.
5. I have some doubts about the recurrence relations in the hyperbolic-tangent magnetic-field case, in particular in Section 5.1, pp. 11–13. Products of two

time series should generally lead to Cauchy convolutions. For example, the coefficient of $v_y(\tau)B_z(\tau)$ should be

$$\sum_{j=0}^i v_{y,j} B_{z,i-j},$$

not simply $v_{y,i}B_{z,i}$ (p. 13). Please carefully check the formulae in this section. If there are errors either in the text or in the code, they should of course be corrected. Moreover, please clarify the derivation of the recurrence for $B_z = \tanh(\gamma y)$.

6. Tethering is important for the reported long-time accuracy. Thus, the role of tethering should be explained more rigorously. Does it preserve the order of the PS method? Can it affect conservation properties or introduce additional errors?
7. The comparison based on identical time steps is useful but not sufficient. Since the PS method has a larger cost per step but can apparently use larger time steps, the relevant comparison should be based on work-precision diagrams. The authors should report runtime versus error for several step sizes or tolerances.
8. RK45 tolerances are changed from one experiment to another. The authors should justify these choices and report the number of function evaluations.
9. The implementation details of the RKG method are insufficient. Since this is an implicit method, the authors should report the nonlinear-solver tolerance, stopping criterion, average number of Newton iterations, maximum number of Newton iterations, and failure criterion. Moreover, the authors should clarify whether any of the standard techniques for efficiently solving the nonlinear systems arising in implicit Runge-Kutta methods were used. For instance, simplified Newton iterations and related implementation strategies for implicit Runge-Kutta schemes are discussed in [Hairer and Wanner, *Solving Ordinary Differential Equations II*].
10. Kinetic-energy conservation is an important diagnostic, but it is not sufficient by itself. A method can conserve energy well and still accumulate phase or trajectory error. The authors should include more systematic trajectory or phase-error diagnostics when a reference solution is available.
11. In the uniform-field case, an exact solution is available. The authors should exploit this, for example by giving work-precision plots for both trajectory error and kinetic-energy error.

12. The wall-clock runtime comparisons may be implementation-dependent. The authors should report other cost indicators, such as number of time steps, number of right-hand-side evaluations, average PS order, and number of nonlinear iterations for RKG.
13. The notation “PS16”, “PS20”, etc. should be clarified.
14. The adaptive truncation criterion should be described more precisely. Using machine epsilon as a stopping tolerance may not be optimal, and in some examples the authors themselves state that lower PS orders would give essentially the same result. This suggests that the truncation strategy is not yet fully optimized.
15. Reproducibility should be improved. The authors should provide pseudocode, or at least a detailed algorithmic description of the PS implementation.

Below, I also list some minor issues that should be reviewed.

- The authors state that the problems considered are not strongly stiff, but later report failures of RK4 and RK45 in long-time simulations. It would be useful to clarify whether these failures are due to stiffness, stability restrictions, accumulated phase/energy errors, or other numerical issues.
- The figure captions are sometimes very long and repeat information already given in the text. They could be shortened.
- The authors should carefully proofread the manuscript. There are several small typos. For example, on p. 6 the text says “an non-indexed symbol”, which should be “a non-indexed symbol”; on p. 7 there is a missing space in “Eqs. 23–25,we obtain”; on p. 12 there appears to be an unresolved or incorrect reference, “Eq. refref:tanh”; on p. 16 the text refers to “Figure 2b” although the discussion concerns Figure 4b; on p. 20 the sentence “where and μ_0 the vacuum permeability” should be corrected; in the caption of Figure 12, on p. 29, the marker size is said to indicate “proton energy”, although the figure concerns electron simulations; finally, around Figure 11 there seems to be an inconsistency between “PS16” in the text and “PS15” in the caption. I suggest a careful revision of the manuscript.